

Rainfall Prediction System

Machine learning model to predict next-day rainfall using historical weather data, built with scikit-learn.

Overview

This project uses real weather station data (temperature, humidity, pressure, cloud cover, wind speed) to predict whether it will rain the next day. The dataset contains ~145,000 records from Australian weather stations.

Approach

- **Data cleaning:** handled missing values using median imputation for numeric features, encoded categorical Yes/No fields
- **Class imbalance:** addressed using `class_weight='balanced'` since "No Rain" days significantly outnumber "Rain" days in the dataset
- **Model:** Random Forest Classifier (200 trees, max depth 12), chosen for its robustness to noisy weather data and ability to rank feature importance
- **Evaluation:** 80/20 train-test split with stratified sampling to preserve class distribution

Results

(Run the notebook to generate your own numbers, then update this section)

- Accuracy: XX.X%
- F1-Score: 0.XX
- Key predictive features: Humidity3pm, Pressure3pm, Cloud3pm (update based on actual output)

Tech Stack

Python, Pandas, NumPy, Scikit-learn, Random Forest

How to Run

1. Open `rainfall_prediction.py` in Google Colab or Jupyter
2. Run all cells — the dataset downloads automatically
3. Review printed accuracy/F1 metrics and feature importance

Future Improvements

- Compare against XGBoost and LightGBM
- Add hyperparameter tuning with GridSearchCV
- Deploy as a simple Streamlit web app for live predictions